

analyses; a detailed linear stability study comparing the various schemes considered in this work is conducted in Section 6; numerical simulations of the SWE on the rotating sphere are performed in Section 7 in order to assess and compare stability, convergence and computational cost properties; and conclusions are presented in Section 8.

2. THE SHALLOW-WATER EQUATIONS ON THE ROTATING SPHERE

The shallow-water equations on the rotating sphere are a popular two-dimensional model in the early stages of atmospheric circulation research since it is simplified w.r.t. more complete, three-dimensional equations but still contains most of the challenges related to the spatial discretization of atmospheric models. Therefore, it allows an easier but insightful study of spatial and temporal discretization schemes for atmospheric models [44]. The equations read

$$\frac{\partial}{\partial t} \mathbf{U} = \mathbf{L}_G(\mathbf{U}) + \mathbf{L}_C(\mathbf{U}) + \mathbf{N}_A(\mathbf{U}) + \mathbf{N}_R(\mathbf{U}) + \mathbf{L}_\nu(\mathbf{U}) + \mathbf{b}, \quad (2.1)$$

where the solution vector $\mathbf{U} = (\Phi, \xi, \delta)^T$ contains the geopotential $\Phi = \bar{\Phi} + \Phi' = gh$, vorticity $\xi := \mathbf{z} \cdot (\nabla \times \mathbf{V})$ and divergence $\delta := \nabla \cdot \mathbf{V}$ fields. Φ, ξ and δ are functions of (λ, μ, t) , where λ is the longitude, $\mu = \sin(\theta)$, θ is the latitude and t is the time. g denotes the gravitational acceleration, h is the fluid depth, $\bar{\Phi}$ and Φ' are respectively the mean geopotential and the geopotential perturbation, $\mathbf{V} := (u, v)^T$ is the horizontal velocity field and $\mathbf{z}$ is the unit vector in the vertical direction. The RHS of (2.1) is decomposed as a sum of linear and nonlinear terms accounting for different physical processes: $\mathbf{L}_G$ accounts for linear processes related to the gravitation; $\mathbf{L}_C$ contains linear rotation-related phenomena, described by the Coriolis parameter $f = 2\Omega \sin \theta$, where Ω is the Earth's angular velocity; $\mathbf{N}_A$ is the nonlinear advection term, $\mathbf{N}_R$ contains the remaining nonlinear terms, $\mathbf{L}_\nu(\mathbf{U})$ is a linear term accounting for an artificial (hyper-)viscosity of order q and coefficient $\nu \geq 0$, with q even, and $\mathbf{b}$ is related to the topography field b . These terms are given by

$$\begin{aligned} \mathbf{L}_G(\mathbf{U}) &= \begin{pmatrix} 0 & 0 & -\bar{\Phi} \\ 0 & 0 & 0 \\ -\nabla^2 & 0 & 0 \end{pmatrix} \mathbf{U}, & \mathbf{L}_C(\mathbf{U}) &= \begin{pmatrix} 0 \\ -\nabla \cdot (f\mathbf{V}) \\ \mathbf{z} \cdot \nabla \times (f\mathbf{V}) \end{pmatrix} \\ \mathbf{N}_A(\mathbf{U}) &= \begin{pmatrix} -\mathbf{V} \cdot \nabla \Phi \\ -\nabla \cdot (\xi \mathbf{V}) \\ -\nabla^2 \left(\frac{\mathbf{V} \cdot \mathbf{V}}{2} \right) + \mathbf{z} \cdot \nabla \times (\xi \mathbf{V}) \end{pmatrix}, & \mathbf{N}_R(\mathbf{U}) &= \begin{pmatrix} -\Phi' \delta \\ 0 \\ 0 \end{pmatrix}, \\ \mathbf{L}_\nu(\mathbf{U}) &= (-1)^{\frac{q}{2}+1} \nu \begin{pmatrix} \nabla^q \Phi' \\ \nabla^q \xi \\ \nabla^q \delta \end{pmatrix}, & \mathbf{b} &= \begin{pmatrix} 0 \\ 0 \\ -g \nabla^2 b \end{pmatrix}. \end{aligned} \quad (2.2)$$

2.1. Spectral discretization

In this work, the spatial dimension is discretized considering a spectral approach using spherical harmonics, which is briefly described below. Details can be found in [39]. Let $Y_{m,n}$ denote the spherical harmonic function of zonal and total wavenumbers m and n , respectively. The spherical harmonics expansion of a given smooth field ψ reads

$$\psi(\lambda, \mu, t) = \sum_{m=-\infty}^{\infty} \sum_{n=|m|}^{\infty} \psi_{m,n}(t) Y_{m,n}(\lambda, \mu) \quad (2.3)$$

where $\psi_{m,n}$ is the expansion coefficient corresponding to the wavenumber pair (m, n) . We consider a triangular truncation with resolution M :

$$\psi(\lambda, \mu, t) = \sum_{m=-M}^M \sum_{n=|m|}^M \psi_{m,n}(t) Y_{m,n}(\lambda, \mu) \quad (2.4)$$

As described in the next section, the use of a spectral discretization based on spherical harmonics, combined with a proper arrangement of the terms of the governing equations, is essential for ensuring an efficient computation of matrix exponentials in both Eulerian and semi-Lagrangian exponential methods.

3. TIME INTEGRATORS

We present in this section the time stepping schemes considered in the work. We begin by briefly presenting the Eulerian exponential integration methods ETD1RK and ETD2RK [8], on which their semi-Lagrangian versions are based. Then, as an introduction to semi-Lagrangian schemes, we present the semi-Lagrangian semi-implicit SETTLS [17], which we will also compare in the numerical experiments. Finally, we present the semi-Lagrangian exponential schemes proposed by [26]. As explained below, we propose a modification to the name of this last class of methods in order to take into account their actual order of convergence.

3.1. Eulerian Runge-Kutta-type exponential integration methods (ETDRK)

Exponential Time Differencing (ETD) schemes are a class of exponential integration methods to solve time-dependent problems of the form

$$\frac{d\mathbf{U}}{dt} = \mathbf{L}\mathbf{U} + \mathbf{N}(\mathbf{U}), \quad \mathbf{U}(t=0) = \mathbf{U}_0, \quad (3.1)$$

which are obtained, *e.g.*, from a spatial discretization of a PDE, where $\mathbf{U} \in \mathbb{R}^N$, $\mathbf{L} \in \mathbb{R}^{N \times N}$ is a linear operator and $\mathbf{N} : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is a nonlinear function. As a common feature, exponential methods integrate the linear term $\mathbf{L}\mathbf{U}$ very accurately (or exactly, down to machine precision, if the exponential terms can be evaluated exactly), by writing its evolution using the exponential of the linear operator, whereas an approximation is proposed to compute the nonlinear term. In the following paragraphs, we briefly describe the main ideas behind exponential methods, more specifically concerning the schemes considered in this work, namely ET1DRK and ETD2RK [8]. We refer the reader to [8, 16] for further details on exponential integration methods and, in particular, to [26] for their derivation through the solution of an integration factor problem, which is also the starting point for the derivation of semi-Lagrangian exponential schemes in Section 3.3..

The exact solution of (3.1) at time $t_{n+1} = t_n + \Delta t$, integrated from $\mathbf{U}(t_n)$, reads

$$\begin{aligned} \mathbf{U}(t_{n+1}) &= P_n^{-1}(t_{n+1})\mathbf{U}(t_n) + P_n^{-1}(t_{n+1}) \int_{t_n}^{t_{n+1}} P_n(s) \mathbf{N}(\mathbf{U}(s)) ds \\ &= e^{\Delta t \mathbf{L}} \mathbf{U}(t_n) + \int_{t_n}^{t_{n+1}} e^{-(s-t_n)\mathbf{L}} \mathbf{N}(\mathbf{U}(s)) ds, \end{aligned} \quad (3.2)$$

which is the well-known variation-of-constants formula, where, assuming that $\mathbf{L}$ does not depend on time, the integration factor

$$P_n(t) = e^{-(t-t_n)\mathbf{L}} \in \mathbb{R}^{N \times N} \quad (3.3)$$

is the solution of the integration factor problem

$$\frac{dP_n}{dt}(t) = -P_n(t)\mathbf{L}, \quad P_n(t_n) = I. \quad (3.4)$$

The first term in (3.2) is the exact integration of the linear term. Different exponential methods can be formulated proposing different approximations to the integral containing the evolution of the nonlinear term. In particular, ETD RK methods are a class of Runge-Kutta type exponential schemes proposed by [8], with improved stability properties w.r.t. previous exponential methods. The first- and second-order ETD RK schemes, named respectively ETD1RK and ETD2RK, read

$$\mathbf{U}_{\text{ETD1RK}}^{n+1} = \varphi_0(\Delta t \mathbf{L}) \mathbf{U}^n + \Delta t \varphi_1(\Delta t \mathbf{L}) \mathbf{N}(\mathbf{U}^n), \quad (3.5a)$$

$$\mathbf{U}_{\text{ETD2RK}}^{n+1} = \mathbf{U}_{\text{ETD1RK}}^{n+1} + \Delta t \varphi_2(\Delta t \mathbf{L}) [\mathbf{N}(\mathbf{U}_{\text{ETD1RK}}^{n+1}) - \mathbf{N}(\mathbf{U}^n)], \quad (3.5b)$$

where

$$\begin{aligned} \varphi_0(z) &= e^z, \\ \varphi_k(z) &= z^{-1} (\varphi_{k-1}(z) - \varphi_{k-1}(0)), \quad k \geq 1 \end{aligned} \quad (3.6)$$

with singularities at $z = 0$ being computed via series expansions in its neighborhood. The functions φ_k are defined both when z is a scalar or a matrix (in particular, we are interested in the case $z = \Delta t \mathbf{L}$). The integrals in the first- and second-order methods are approximated by considering respectively constant and linear approximations to the nonlinear function along $[t_n, t_{n+1}]$:

$$\begin{aligned} \mathbf{N}(\mathbf{U}(s)) &\approx \mathbf{N}(\mathbf{U}(t_n)) \\ \mathbf{N}(\mathbf{U}(s)) &\approx \mathbf{N}(\mathbf{U}(t_n)) + \frac{s - t_n}{\Delta t} [\mathbf{N}(\mathbf{U}_{\text{ETD1RK}}(t_{n+1})) - \mathbf{N}(\mathbf{U}(t_n))] \end{aligned} \quad (3.7)$$

As mentioned above, these schemes solve the linear term exactly, if the exponential terms can be computed exactly (down to machine precision), which is the case, *e.g.*, of scalar equations. However, in the case of systems of equations, this is possible only if the matrix exponentials $e^z = \sum_{k=0}^{\infty} z^k/k!$ can be computed exactly. In general, the matrix exponential must be evaluated approximately, and the efficiency of traditional algorithms limits their practical implementation in geophysical contexts [13, 30] thus requiring the development of more efficient algorithms. Some examples, that have been applied to atmospheric modeling, are Krylov subspace methods with incomplete orthogonalization [13] and rational approximation to exponential integrators [14, 33]. Nevertheless, if z is diagonalizable, then the matrix exponential can be computed exactly [26]: if $z = Q\Lambda Q^{-1}$, where the columns of Q are the eigenvectors of z and Λ is a diagonal matrix containing the respective eigenvalues, then

$$\varphi_k(z) = Q \varphi_k(\Lambda) Q^{-1}, \quad k \geq 0, \quad (3.8)$$

where $\varphi_k(\Lambda)$ can be computed elementwise since Λ is diagonal.

We make use of this property to evaluate the matrix exponentials in a relatively efficient manner. We remark that, in the case of the SWE on the rotating sphere, the linear term $\mathbf{L}_G + \mathbf{L}_C$ is not diagonalizable, since the Coriolis parameter is spatially dependent. In the case where f is constant (*e.g.*, in the so-called f -plane approximation), $\mathbf{L}_G + \mathbf{L}_C$ is diagonalizable, which is taken into